

IF

THE UNIVERSE IS ALL THERE IS, HOW CAN IT EXPAND?

When a glowing object moves away from us, its light is known to become redder than usual. Measurements of galaxy clusters show that their light is reddened, so they must be moving away. The universe has been expanding since the big bang. This is not because extra matter is appearing but because galaxy clusters are getting farther apart. Maybe there is extra room in the universe if there is space for galaxies to spread farther apart, but exactly how much room is left we do not know. The universe cannot be expanding into anything other than itself, for whatever it is expanding into is part of the universe, as there is nothing other than the universe.

WHICH

BRIGHT SPARK FIGURED OUT THAT THE UNIVERSE IS EXPANDING?

In the 1910s, American astronomer Vesto Slipher discovered that the light from some galaxies was redder than others. By the 1920s, American astronomer Edwin Hubble had found a way to measure the distances of galaxies, so he and fellow astronomer Milton Humason compared the levels of reddening with these distances. They discovered that the more distant galaxies were moving away faster, which suggested that the universe was expanding.

WHAT

SIZE IS THE UNIVERSE TODAY?

Measuring units, such as miles and kilometers, are regularly used on Earth, but they are too small to describe the mind-boggling distances beyond the solar system. Light travels at an amazing 186,282 miles (299,792 km) per second. A light-year is the distance that light travels in one year—about 6 trillion miles (10 trillion km). The detectable universe is at least 90 billion light-years across. Nothing moves faster than light, but distances in the universe are so great that it can take years for the light to reach us.

HOW

WILL THE UNIVERSE END?

Scientists think the universe will end in one of three ways—either the galaxies, stars, and atoms in the universe will eventually rip themselves apart or the universe will expand forever, gradually cooling until it is completely dark and dead. It is also possible that the universe will stop expanding and crash in on itself. These three theories are called the big rip, big chill, and big crunch.

